



BHABHA UNIVERSITY
SCHEME OF EXAMINATION

BACHELOR OF TECHNOLOGY (BTECH) III SEM

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

SESSION 2019-20 W.E.F AUGUST 2019

S.NO	SUBJECT CODE	SUBJECT NAME	MAXIMUM MARKS ALLOTTED						TOTAL MARKS	HOURS/ WEEK			TOTAL CREDITS	REMARKS
			THEORY			PRACTICALS				L	T	P		
			END SEM	MID SEM	QUIZ ASSIGN - MENTS	END SEM	LAB WORK	ASSIGN - MENTS QUIZ						
1	CH-31	ENERGY & ENVIROMENTAL ENGINEERING	70	20	10	—	—	—	100	3	1	—	4	
2	EC-31	DIGITAL CIRCUITS & SYSTEM	70	20	10	30	10	10	150	3	1	2	5	
3	EC-32	NETWORK ANALYSIS	70	20	10	30	10	10	150	3	-	2	4	
4	EC-33	ELECTRONIC DEVICES & CIRCUITS	70	20	10	30	10	10	150	3	1	2	5	
5	EC-34	MEASUREMENTS AND INSTRUMENTATION	70	20	10	30	10	10	150	3	1	2	5	
6	EC-35	NSS/NCC/SOCIAL WORKS (INTERNAL ASSESSMENTS)	—	—	—	—	—	—	50	—	—	1	1	TOTAL MARKS
TOTAL			350	100	50	120	40	40	750	15	4	5	24	750

750



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DEPARTMENT OF ELECTRONICS AND COMMUNICATION

B.TECH III SEM

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

SYLLABUS

CH-31 ENERGY & ENVIRONMENTAL ENGINEERING

UNIT –I

Energy- Sources of Energy : Renewable & Non Renewable, Fossil fuel, Biomass Geothermal, Hydrogen, Solar, Wind, hydal, nuclear sources.

UNIT –II

Ecosystem – Segments of Environment: Atmosphere, hydrosphere, Lithosphere, biosphere. Cycles in Ecosystem – Water, Carbon, Nitrogen. Biodiversity: Threats and conservation,

UNIT –III

Air Pollution & Sound Pollution -Air Pollution: Air pollutants, classification, (Primary & secondary Pollutants) Adverse effects of pollutants. Causes of Air pollution chemical, photochemical, Green house effect, ozone layer depletion, acid Rain. Sound Pollution: Causes, controlling measures, measurement of sound pollution (decibel), Industrial and non – industrial.

UNIT –IV

Water Pollution– Water Pollution: Pollutants in water, adverse effects. Treatment of Domestic & Industrial water effluent. Soil Pollution – Soil Profile, Pollutants in soil, their adverse effects, controlling measures.

UNIT –V

Society, Ethics & Human values– Impact of waste on society. Solid waste management (Nuclear, Thermal, Plastic, medical, Agriculture, domestic and e- waste). Ethics and moral values, ethical situations, objectives of ethics and its study . Preliminary studies regarding Environmental Protection Acts , introduction to value education, self exploration, sanyam & swasthya.

References:

1. Harris, CE, Prichard MS, Rabin's MJ, "Engineering Ethics"; Cengage Pub.
2. Rana SVS ; "Essentials of Ecology and Environment"; PHI Pub.
3. Raynold, GW "Ethics in information Technology"; Cengage.
4. Svakumar; Energy Environment & Ethics in society; TMH
5. AK De "Environmental Chemistry"; New Age Int. Publ.
6. BK Sharma, "Environmental Chemistry" ; Goel Publ. House

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION

B.TECH III SEM

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

SYLLABUS

EC-31 DIGITAL CIRCUITS & SYSTEM

UNIT1

Review of Logic gates and binary operations- AND, OR, NOT, NAND, NOR, Exclusive-OR and Exclusive-NOR Implementations of Logic Functions using gates, NAND-NOR implementations – Multi level gate implementations- Multi output gate implementations. Introduction to number systems and binary operations.

Boolean postulates and laws – De-Morgan's Theorem - Principle of Duality, Boolean function, Canonical and standard forms, Minimization of Boolean functions, Minterm, Maxterm, Sum of Products (SOP), Product of Sums (POS), Karnaugh map Minimization, Don't care conditions, Quine-McCluskey method of minimization.

UNIT-II

Combinational logic circuits: Half adder – Full Adder – Half subtractor - Full subtractor – Parallel binary adder, parallel binary Subtractor – Fast Adder - Carry Look Ahead adder – Serial. Adder/Subtractor - BCD adder – Binary Multiplier – Binary Divider - Multiplexer/De-multiplexer – decoder - encoder – parity checker – parity generators – code converters - Magnitude Comparator.

UNIT-III

Sequential logic circuits: Latches, Flip-flops - SR, JK, D, T, and Master-Slave, Characteristic table and equation – Application table – Edge triggering – Level Triggering – Realization of one flip flop using other flip flops – serial Adder/subtractor.

UNIT-IV

Registers and Counters: Asynchronous Ripple or serial counter. Asynchronous Up/Down counter - Synchronous counters – Synchronous Up/Down counters – Programmable counters – Design of Synchronous counters: state diagram-State table – State minimization – State assignment - Excitation table and maps-Circuit. Implementation - Modulo-n counter, Registers – shift registers - Universal shift registers. Shift register counters – Ring counter – Shift counters - Sequence generators.

UNIT-V.

Logic Families: Introduction to different logic families and their characteristics, RTL, DTL, TTL, ECL, IIL, TTL inverter – circuit description and operation, CMOS inverter – circuit description and operation, other TTL and CMOS gates, Memories – ROM - ROM organization - PROM – EPROM – EEPROM – EAPROM, RAM – RAM organization Static RAM, Dynamic RAM, Programmable Logic Array (PLA) - Programmable Array Logic (PAL)



BHABHA UNIVERSITY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION

B.TECH III SEM

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

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TEXT BOOKS

1. M. Morris Mano, Digital Design, 3rd Edition, Prentice Hall of India Pvt. Ltd., 2003 Pearson Education (Singapore) Pvt. Ltd., New Delhi, 2003.
2. Donald P. Leach and Albert Paul Malvino, Digital Principles and Applications, 6th Edition, TMH, 2003.
3. <http://www.nptelvideos.in/2012/12/digital-circuits-and-systems.html>

Graphical Programming using Lab

1. To verify the truth table of all basic logic gates and to implement all gate using universal gate.
2. Design of 3 bit Adders (CLA, CSA, CMA, Parallel adders)
3. Design of Binary Subtractors
4. Design of Encoder (8X3), Encoder (3X8)
5. Design of Multiplexer (8X1), (16X1) and De-multiplexer (1X8), (1X16)
6. Design of code converters & Comparator
7. Design of FF (SR, D, T, JK, and Master Slave with delays)
8. Design of registers using latches and flip-flops
9. Designing a synchronous and asynchronous counters.

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION

B.TECH III SEM

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

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EC-32 NETWORK ANALYSIS

UNIT-I

Introduction to circuit theory: basic circuit element R,L,C and their characteristics in terms of linearity & time dependant nature, voltage & current sources, controlled & uncontrolled sources KCL and KVL analysis, Concept of phasor & vector, impedance & admittance, Nodal & mesh analysis, analysis of magnetically coupled circuits. Dot convention, coupling coefficient, tuned circuits, Series & parallel resonance.

UNIT-II

Network Graph theory: Concept of Network graph, Tree, Tree branch & link, Incidence matrix, cut set and tie set matrices, dual networks.

UNIT-III

Network Theorems: Thevenins & Norton's, Super positions, Reciprocity, Compensation, Substitution, Maximum power transfer, and Millman's theorem, Tellegen's theorem, problems with dependent & independent sources.

UNIT-IV

Transient analysis: Transients in RL, RC&RLC Circuits, initial& final conditions, time constants. Steady state analysis Laplace transform: solution of Integro-differential equations, transform of waveform synthesized with step ramp, Gate and sinusoidal functions, Initial & final value theorem, Network Theorems in transform domain.

UNIT-V

Two port parameters: Z, Y, ABCD, Hybrid parameters, their inverse & image parameters, relationship between parameters, Interconnection of two ports networks, Reciprocity and Symmetry in all parameter.



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DEPARTMENT OF ELECTRONICS AND COMMUNICATION

B.TECH III SEM

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

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EXPERIMENTS LIST:-

1. To Verify Thevenin Theorem and Superposition Theorem.
2. To Verify Reciprocity Theorem and Millman's Theorem.
3. To Verify Maximum Power Transfer Theorem.
4. To Determine Open Circuit and Short Circuit parameters of a Two Port Network.
5. To Determine A,B, C, D parameters of a Two Port Network.
6. To determine h parameters of a Two Port Network.
7. To Find Frequency Response of RLC Series Circuit RLC parallel Circuit and determine resonance and 3dB frequencies.
8. To determine charging and discharging times of Capacitors.

TEXT BOOKS:-

1. M.E. Van Valkenburg, Network Analysis, (Pearson)
2. S P Ghosh A K Chakraborty Network Analysis & Synth. (MGH).
3. <http://www.nptelvideos.in/2012/11/networks-and-systems.html>

REFERENCE:-

1. Sudhakar-Circuit Network Analysis & Synth(TMh).
2. J. David Irwin Engineering Circuit analysis tenth edition, Wiley india.
3. Kuo- Network Analysis & Synthesis, Wiley India.
4. Robert L Boylestad introductory Circuit analysis, Pearson
5. Smarajit Ghosh, NETWORK THEORY: ANALYSIS AND SYNTHESIS (PHI).
6. Roy Choudhary D; Network and systems; New Age Pub.
7. Bhattacharya and Singh- Network Analysis & Synth (Pearson). 10.Study of types of memories.



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EC-33 ELECTRONIC DEVICES & CIRCUITS

UNIT-I

Introduction to semiconductor physics: insulator, conductor, semiconductor and semiconductor types. Drift and diffusion carries, Hall Effects.

Review of PN junction diode: PN junction diode in forward and reverse bias, temperature dependence of V-I characteristics, diode resistances, diode junction capacitance. Types of diodes: Zener Diode, Varactor Diode, Tunnel Diode, PIN Diode, Schottky Diode, LED and Photo Diodes, Switching characteristics of diode.

UNIT-II

Bipolar junction transistor - Construction, basic operation, current components and equations, CB, CE and CC- configuration, input and output characteristics, Early effect, Region of operations: active, cut-off and saturation region. BJT as an amplifier. Ebers-Moll model, Power dissipation in transistor (P_d , max rating), Photo transistor.

Transistor biasing circuits and analysis: Introduction, various biasing methods: Fixed bias, Self bias, Voltage Divider bias, Collector to base bias, Load-line analysis: DC and AC analysis, Operating Point and Bias Stabilization and Thermal Runaway. Transistor as a switch.

UNIT-III

Small Signal analysis: Small signal Amplifier, Amplifier Bandwidth, Hybrid model, analysis of transistor amplifier using h-parameter, Multistage Amplifier: Cascading amplifier, Boot-strapping Technique, Darlington amplifier and cas-code amplifier, Coupling methods in multistage amplifier, Low and high frequency response, Hybrid π model, Current Mirror circuits.

UNIT-IV

Large Signal analysis and Power Amplifiers: Class A, Class B, Class AB, Class C, Class D, Transformer coupled and Push-Pull amplifier.

UNIT-V

FET construction- JFET: Construction, n-channel and p-channel, transfer and drain characteristics, parameters, Equivalent model and voltage gain, analysis of FET in CG, CS and CD configuration. Enhancement and Depletion MOSFET drain and transfer Characteristics.

Uni-junction Transistor (UJT) and Thyristors: UJT: Principle of operation, characteristics, UJT relaxation oscillator, PNP Diode and its characteristics, Silicon controlled rectifier: V-I characteristics, DIAC and TRIAC, Thyristors parameters and applications.



BHABHA UNIVERSITY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION

B.TECH III SEM

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

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LIST OF EXPERIMENTS:

1. To determine and analyze the V-I characteristics of PN Junction diode and Zener diode.
2. To determine input and output characteristics of transistor amplifiers in CE, CB & CC configurations.
3. To determine the frequency response of transistor CE amplifier, direct coupled and RC coupled amplifier.
4. To determine characteristics of UJT as relaxation Oscillator.
5. To determine Drain and Transfer Characteristics of JFET Amplifier.
6. To determine Drain and Transfer Characteristics of MOSFET Amplifier.
7. To determine characteristics of class A and B power amplifiers.
8. To determine characteristics of class C and AB power amplifiers.

REFERENCES:

1. **Sedra and Smith: Microelectronics, Oxford Press.**
2. Anil K. Maini, Varsha Agarwal: Electronic Devices and Circuits, Wiley Publications.
3. Rashid: Electronic Devices and Circuits, Cengage learning.
4. **Donald A Neamen: Electronic Circuits Analysis and Design, TMH**
5. **Salivahanan: Electronic Circuits Analysis and Design, TMH**
6. Mottershead: Electronic Devices and Circuits an introduction, PHI
7. Kumar and Jain: Electronic Devices and Circuits, PHI.
8. David A. Bell Electronic Devices and Circuits Oxford University press.





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DEPARTMENT OF ELECTRONICS AND COMMUNICATION

B.TECH III SEM

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

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EC-34 MEASUREMENTS AND INSTRUMENTATION

UNIT-I

Accuracy and Precision, Sensitivity, Linearity, Resolution, Hysteresis, Loading Effect. Measurements of Current, Voltage, Power and Impedance: DC and AC Ammeter, DC Voltmeter- Chopper type and solid-state, AC voltmeter using Rectifier. Average, RMS, Peak responding voltmeters, Multi-meter, Power meter, Bolometer and Calorimeter.

UNIT-II

Different parts of CRO, Block diagram, Electrostatic focusing, Electrostatic deflection, Post deflection acceleration. Screen for CRTs, Graticules, Vertical and Horizontal deflection system, Time base circuit, Oscilloscope Probes, Applications of CRO, Special purpose CROs- Multi input, Dual trace, Dual beam, Sampling, Storage (Analog and Digital) Oscilloscope.

UNIT-III

Maxwell's bridge (Inductance and Inductance-Capacitance), Hay's bridge, Schering bridge (High voltage and Relative permittivity), Wein bridge. Impedance measurement by Q-meter.

Non-Electrical Quantities (Transducer): Classification of Transducers, Strain gauge, Displacement Transducer- Linear Variable Differential Transformer (LVDT) and Rotary Variable Differential Transformer (RVDT), Temperature Transducer- Resistance Temperature Detector (RTD), Thermistor, Thermocouple, Piezo-electric transducer, Optical Transducer- Photo emissive, Photo conductive, Photo voltaic, Photo-diode, Photo Transistor

UNIT-IV

Signal and Function Generators, Sweep Frequency Generator, Pulse and Square Wave Generator, Beat Frequency Oscillator, Digital display system and indicators, Classification of Displays, Display devices: Light Emitting diodes (LED) and Liquid Crystal Display(LCD).

UNIT-V

Advantages of Digital Instrument over Analog Instrument, Digital-to-analog conversion (DAC) - Variable resistive type, R-2R ladder Type, Binary ladder, Weighted converter using Op-amp and transistor, Practical DAC. Analog-to- digital Conversion (ADC) -Ramp Technique, Dual Slope Integrating Type, Integrating Type (voltage to frequency), Successive Approximations. Digital voltmeters and multi-meters, Resolution and sensitivity of digital multi-meter.



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B.TECH III SEM

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

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List of Experiments:

1. Study of Cathode Ray Oscilloscope and Function Generator.
2. Study of displacement measurement by LVDT.
3. Force measurement by strain gauge.
4. Measurement of Capacitor using Q-meter.
5. Measurement of Self-induction using Q-meter.
6. Temperature measurement by thermistor, RTD and thermocouple.
7. Study of optical Transducers: Photo conductive, Photo voltaic, Photo-diode, Photo-Transistor.
8. Design of digital to analog converter, R-2R ladder Type and analysis of its characteristics.

TEXTBOOKS

1. H.S. Kalsi: Electronics Instrumentation, TMH
2. A.K. Sawhney: Instrumentation and Measurements, Dhanpat Rai and Co. 1. Oliver: electronic Measurements introduction TMH
2. Helfric and Cooper: Modern Electronic Instrumentation and Measurement Techniques, Pearson.
3. Ghosh: introduction to measurements and instrumentation 4th edition PHI
4. Bell: electronic Instrumentation and Measurement oxford press.
5. Banerjee: electrical and electronics Measurement 2nd PHI.
6. Anand: electronics and Instrumentation technology, PHI.

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